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Shahzeb Yousuf – Data Scientist & ML Consultant

Data Scientist and ML Consultant with extensive experience in architecting and deploying data-driven solutions. Proven track record in leading teams in developing high-impact AI systems, including Large Language Models (LLMs) and advanced predictive pipelines. Consulted as a Data Science subject matter expert for the DHA, DoH, various Health insurance companies, Rasan IT, NMC Healthcare and ENEC (Nuclear energy).

Skills:

- **Power BI/Qlik/Tableau** – Dash boarding **(9Yrs)**
- **SQL** – Data Querying and wrangling **(9Yrs)**
- **Project / Stakeholder Management** – **(9Yrs)**
- **AI** – LLM's, NLP, NN's etc **(5Yrs)**
- **R** – Statistical Analysis / Data Visualization **(9Yrs)**
- **Python** – Data processing, cleaning and ML **(9Yrs)**
- **ML** – NNs, Bayesian Inference, MCMC etc **(9Yrs)**
- **Data Engineering** – Databricks, Snowflake etc **(8Yrs)**

Experience:

CURRENT (Details Confidential)

Data Scientist & Data Engineer / Emirates Nuclear Energy Company, Abu Dhabi

NOV 2024 – OCT 2025

Principal Data Scientist / NMC Healthcare, Abu Dhabi

Lead the team in terms of data science. Applied my skills and knowledge in machine learning, data engineering, and healthcare to standardize data sets into unified formats, apply advanced machine learning methods for various sorts of healthcare data. Carried out a large scale data migration project to a cloud, Snowflake.

JUNE 2024 – NOV 2024 (Contract Based 6 Months)

Principal Data Scientist / Rasan Information Technology, Dubai

As the Principal Data Scientist, I applied my skills and knowledge in machine learning, data analysis, and actuarial science to provide solutions for clients in the GCC region. Developing a solution that enables the local execution of large language models (LLMs like Chat GPT) for natural language processing and image processing tasks.

MAY 2023 – MAY 2024

Data Science & Actuarial Consultant / Milliman ME, Dubai

Various data analyses and reports for self-insured Health government programs of the state of Abu Dhabi and private insurers. Experience includes medical pricing, data analysis and reconciliation of health insurance data, DRG data analysis, revenue analysis, electronic audits of health data, ML model to predict and model actuarial problems, etc.

JULY 2019 – APR 2023

Data Science & Actuarial Consultant / Taha Actuaries and Consultants, Dubai

Various data analyses and reports for self-insured Health government programs for the state of Dubai.
Experience includes Utilization Reports, Cost Driver Analyses, tariff negotiations with providers, actuarial reserving and pricing, DRG data analysis, and applied various machine learning methods to actuarial problems. ML methods include, advanced GLM methods for use in reserving, Change Point Detection, ARIMA models, and Logistic regression.

Older Experiences

JAN 2018 – AUG 2018 - Actuarial Analyst (Intern) / Lux Actuaries and Consultants, Dubai

Education and Qualifications:

M.Sc. Data Analytics (Underway) / University of Cumbria
B.Sc. Actuarial Science (and) Financial Mathematics / KFUPM
Certified Machine Learning Professional / London International Studies and Research Center
Associate of the Society of Actuaries (Partway) / Society of Actuaries
AI & ML Engineering Certification / DELL Technologies

Data Portfolio – Projects Led and Delivered

AI/ML: Change Point Detection
Developed an ML pipeline using Bayesian inference and Markov Chain Monte Carlo (MCMC) to detect maintained changes in healthcare service costs from large datasets.

AI/ML: Hospital Readmissions
Created an ML pipeline with Generalized Linear Models to predict readmissions, aiding investigations into medical complications or negligence

AI/ML: Credit default with Deep Learning
Led a team to victory in a national Fintech competition in KSA in 2018 with a neural network model predicting individual credit default risks.

AI/ML: NLP and Image processing
Developed a model that enables the local execution of large language models (LLMs) for natural language processing and image processing tasks.

Geospatial Analysis
Created a model that analyses the geo-coordinates of various insurance claims to identify patterns in customer behavior.

AI/ML: Natural Language Query Interface
Developed a application enabling users to query a PostgreSQL database using natural language. The tool leverages Large Language Models (OpenAI or local Ollama) to translate user questions into insights.

Data pipelines and ETL processes
Engineered complex data pipelines for health data, standardizing incoming data for analytics and ML applications.

Data system migration
Transitioned a data pipeline of the largest healthcare provider in the UAE from a legacy system to Snowflake, adapting code and data to new syntax and platforms.

Automated reporting dashboards
Constructed and managed automated dashboards extracting data from SQL warehouses for non-technical stakeholders, focusing on health utilization and cost.

AI/ML: Various other ML models
Developed and deployed various models including PCA, ARIMA, and predictive GLM for insurance based purposes.

Data Acquisition (Public sources)
Developed a pipeline to extract/scrape data from various public sources including financial data, item prices, vehicle details, and much more.

ML: Actuarial Claims Reserving
Developed a application for actuarial analysis using the Mack Chain Ladder method. The tool allows users to upload medical claims data, calculates ultimate claims and IBNR reserves.

Data Portfolio – Projects Led and Delivered

AI/ML: Agentic Market Intel and Finance

Developed an end-to-end market intelligence pipeline using Google Custom Search API and web scraping to aggregate global uranium industry data, utilizing LLMs. Combined with a engineered a quantitative financial suite in Python.

AI/ML: LLM Driven Sentiment Alpha

Developed a medium-frequency algorithmic strategy that leverages Large Language Models to perform real time sentiment analysis on financial news and social feeds, generating predictive signals for mid-term asset valuation.

Finance/ML: HFT Probabilistic Tool

Developed a model for predicting buy/sell probabilities at micro-scale intervals using neural networks.
Note: Personal experience/project only.

Finance: Insider Trading Analysis

Developed a data pipeline that scrapes and filters S&P 500 insider trading filings, isolating high-conviction transactions $1M. With is we systematically isolate high-conviction financial signals.

AI: Theoretical basis for RNN In C++

Engineered a zero-dependency Recurrent Neural Network (RNN) in C++17, implementing core sequential logic including autoregressive generation, weight initialization, and numerical stability kernels for Softmax and ReLU activations.

Finance: Systematic Volatility Harvesting

Engineered a path-dependent trading framework that cycles between cash-secured puts and covered calls to capture Theta decay.

AI/ML: Nuclear procedures adv search

Developed an AI-powered semantic search platform using a Retrieval-Augmented Generation (RAG) architecture to streamline the retrieval of critical nuclear regulatory information from extensive libraries of NRC Technical Letters.

Research Papers

RNN's for Financial Market Regime Detection / MSc – Research Paper (Underway)

Currently researching and evaluating a vanilla RNN system designed by myself in C++ designed to process multi-dimensional market embeddings for next-day prediction and unsupervised regime detection. The research demonstrates how a network's internal hidden states can naturally organize into clusters corresponding to distinct financial regimes, such as bullish momentum or high-volatility transitions.

Key Technical Contributions:

- Custom C++ Implementation: Built the RNN engine from scratch to ensure full computational traceability, exporting every matrix operation and intermediate vector to CSV for manual verification.
- Market Vector Embeddings: Engineered 35–50 dimensional daily vectors capturing price returns, volatility, sector performance, and macroeconomic signals.
- Interpretability Framework: Applied k-means clustering and dimensionality reduction (PCA, t-SNE) to hidden state trajectories to identify emergent market regimes without explicit labels.
- Anomaly Detection: Developed a protocol using Euclidean distance and rolling error means to identify significant market shocks and geopolitical events.